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Antiepileptic drug withdrawal in juvenile myoclonic epilepsy

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Objectives: Withdrawal of antiepileptic drugs (AEDs) has been discouraged in juvenile myoclonic epilepsy (JME). However, impulsivity as a consequence of executive dysfunction in JME may influence treatment adherence. The aim of the present study was to assess how common withdrawal of AEDs is in a large and representative JME group.**Materials and methods:** Patients with genetic generalized epilepsy (GGE) were identified through a retrospective search of medical records at Drammen Hospital, Norway, and invited to a clinical interview. Information related to AED withdrawal was analyzed in those classified as JME.**Results:** A total of 132 patients with GGE were interviewed (87 JME). Thirty-five patients with JME (40%) discontinued AEDs, of which 74% did so without consulting a doctor. The rate of self-withdrawal was significantly higher in JME than in other types of GGE. Having a parent with psychosocial difficulties was significantly over-represented in the JME self-withdrawal group. Twelve of those who discontinued AEDs (34%) were free from generalized tonic-clonic seizures (GTCS) and without antiepileptic drugs >1 year. All but one of them withdrew AEDs without consulting a doctor. Age at first motor seizure was significantly higher in those with a favorable outcome of AED withdrawal.**Conclusions:** Self-withdrawal of AEDs is common in JME, especially in those with troublesome conditions at home. However, about 1/3 may remain free from GTCS without AEDs. The findings indicate a need for a stronger follow-up with appropriate information about the prognosis of the disorder.

KEYWORDS

adherence, antiepileptic drugs, juvenile myoclonic epilepsy, withdrawal

1 | INTRODUCTION

Juvenile myoclonic epilepsy (JME) is the most common type of epilepsy affecting adolescents, and it is hallmarked by myoclonic jerks, predominantly after awakening, aggravated by sleep deprivation and stress.¹ The majority of patients have occasional generalized tonic-clonic seizures (GTCS), and about one-third have absence seizures.²

The antiepileptic drugs (AEDs) most commonly used in the treatment of JME are valproate, lamotrigine, levetiracetam, topiramate, and zonisamide.³

Juvenile myoclonic epilepsy was initially considered a manageable type of epilepsy, with intellectual abilities within the normal range, normal neurologic examination, normal magnetic resonance imaging of the brain, and a favorable response to treatment in the majority of patients.^{4,5} However, the rate of relapse upon withdrawal of AEDs was high, and the general advice was that these patients ought to continue medication throughout life.⁵⁻⁷ Smaller, but more

Cecilie Johannessen Landmark and Jeanette Koht contributed equally and are joint last authors.

recent, retrospective studies support this, with seizure relapse in eight of ten and 17/17 JME patients attempting AED withdrawal.^{8,9} Consequently, people with JME have been excluded from prospective AED withdrawal trials.^{10,11}

Lifelong treatment of a disorder starting in youth means exposure to a considerable burden of medication, including short-term adverse effects, which generally subside when treatment is ended, and long-term adverse effects. Long-term adverse effects may persist for months or years after withdrawal of the causative agent. Moreover, AEDs is a challenging group of drugs to be managed in mono- or polytherapy due to pronounced pharmacokinetic variability and numerous pharmacokinetic interactions.^{12,13} It is not certain, however, that all people with JME need treatment with AEDs throughout life. Some are able to control seizures by avoiding trigger factors like sleep deprivation and stress.¹⁴ Several long-term follow-up studies of JME state that some patients may discontinue AED treatment and remain seizure-free.¹⁵⁻²¹ Discrepancies between medical advice and the patient's experience or ability to follow such advice may lead to distrust, and it is likely that people with JME experience challenges with adherence to treatment as a potential consequence of executive dysfunction and impulsive decision making.²²⁻²⁴ These issues are probably overlooked in clinical practice, as patients distrusting their doctor will most likely not return to the hospital for follow-up. The aim of the present study was to assess the magnitude of AED withdrawal in a large and representative group of patients with JME.

2 | METHODS

2.1 | Study area

Recruitment of patients was based on a search of medical records at Drammen Hospital in the time period 1999-2013. All medical records containing an International Classification of Diseases, 10th Revision (ICD-10) code of epilepsy (G40) were reviewed.²⁵ Drammen Hospital serves the inhabitants of Buskerud county and four neighboring municipalities, that is, a population of 477 000 people, or 9.1% of Norway's total population.²⁶ Buskerud county has one department of neurology, one department of pediatrics, and one electroencephalogram (EEG) laboratory, all located at Drammen Hospital. There is one private neurologist, and two private pediatricians in the county, who will refer patients to Drammen Hospital when in need of an EEG recording. EEG recordings and consultations with a specialist are standard procedure when diagnosing epilepsy in Norway.²⁵ Thus, we hypothesized that the vast majority of patients with epilepsy in our region visited Drammen Hospital at least once and would be registered in our records.

2.2 | Participants

As the diagnosis of JME can be highly unreliable based on medical records only,¹ all patients with genetic generalized epilepsy (GGE) aged 14-40 years were invited to a clinical interview at Drammen

Hospital. Patients older than 40 years were not contacted, as including older age-groups could have biased our results toward more treatment-resistant epilepsy. We assumed that several of those diagnosed prior to 1999 who responded well to treatment did not return to the hospital and would not be identified in our search of medical records from 1999 to 2013. Patients with childhood absence epilepsy (CAE) who did not use AEDs and were seizure-free for more than 1 year were not contacted. If a patient was younger than 18 years, the parents were contacted. After finalizing the search of medical records, patients with GGE aged 14-40 years were consecutively recruited from the EEG laboratory at Drammen Hospital. Those who declined a hospital visit were offered a home visit.

2.3 | Clinical interview

The clinical interviews were conducted between November 2016 and February 2018 at Drammen Hospital or in the participant's home. A semistructured questionnaire designed for the purpose of this study was used. The questionnaire contained sections on background and medical history, AED treatment, and AED withdrawal. Additionally, the Hospital Anxiety and Depression Scale (HADS) and the Barratt Impulsiveness Scale (BIS) were administered.^{27,28} HADS is scored on a 0-42 point scale, with higher scores indicating more anxiety and depression. Total BIS score may range from 30 to 120, with higher scores indicating more impulsive behavior. The interviews were organized for research purpose only, independent of regular clinical follow-up.

2.4 | Classification and definitions

As suggested by the International League against Epilepsy (ILAE), GGE included the following electroclinical epilepsy syndromes: JME, CAE, juvenile absence epilepsy (JAE), and epilepsy with generalized tonic-clonic seizures only.²⁹ The definition of JME was based on the consensus on diagnosing and management of JME issued in 2013.² Myoclonic jerks had to be the dominating seizure type.¹ Patients were classified as JME or non-JME based on information from the clinical interview, medical records, and EEG recordings. Generalized spike or polyspike and wave activity had to be present on at least one EEG recording, and EEG background activity had to be normal.² Polytherapy was defined as two or more AEDs used concomitantly. Self-withdrawal was defined as intentional discontinuation of antiepileptic medication without consulting a doctor.

2.5 | Statistical methods

All statistical analyses were performed using the Statistical Package for the Social Sciences (SPSS) software, version 23. Variables potentially influencing AED self-withdrawal were analyzed one by one, as were variables potentially influencing the outcome of withdrawal. Student's *t* tests were used for comparison of continuous variables, and chi-square tests were used for comparison of categorical variables, with Yate's continuity correction

for 2x2 tables. When expected cell count was less than five in any cell, Fisher's exact probability test was used. *P*-values ≤ 0.05 were considered significant.

2.6 | Ethics

The study was approved by the Regional Committee for Medical Research Ethics, South East Norway (ethical agreement no. 2013/1027) and by the data protection officer of Drammen Hospital. Written informed consent was obtained from all study participants, and from parents if the participant was younger than 18 years.

3 | RESULTS

3.1 | Participants

We contacted 205 patients with GGE, of which 141 (69%) agreed to participate in the study. Eighty-seven patients (62%) were classified as JME. Forty-five (32%) were classified as other types of GGE. Nine patients were excluded as they could not be classified as GGE according to the inclusion criteria. Demographic and clinical characteristics of the patients with JME and other types of GGE are summarized in Table 1. Antiepileptic drugs used in the patients with JME are presented in Figure 1. Eighteen patients with JME had been off AED medication for >1 year. Two of them (11%) had a GTCS within the last year. In comparison, 18 of the 74 patients using AEDs (24%) had a GTCS within the last year. Fifteen patients not using AEDs (83%) had myoclonic jerks within the last year, compared to 48 (65%) of those using AEDs. The differences were not statistically significant.

3.2 | Rate of antiepileptic drug withdrawal

Thirty-five patients with JME (40%) had a history of withdrawing AED treatment, of which 26 (74%) did so without consulting a doctor. In the group with other types of GGE, 24 patients (53%) withdrew AED, of which ten (41%) did so without consulting a doctor. The rate of self-withdrawal was significantly higher in the JME group ($P = 0.024$). Factors potentially influencing self-withdrawal in JME are presented in Table 2. A parent with psychosocial difficulties like addiction or violent behavior (as reported by the patient) was significantly more common in the group with a history of AED self-withdrawal ($P = 0.006$). Sixteen of those who self-withdrew AEDs (62%) did so within 2 years of their last GTCS.

3.3 | Outcome of antiepileptic drug withdrawal

The outcomes of AED withdrawal in 35 patients with JME are presented in Figure 2. Twelve patients (34%) experienced no relapse of GTCS and did not restart AED therapy. All but one of them self-withdrew their AEDs. Factors potentially influencing outcome of AED withdrawal are presented in Table 3. Age at first motor seizure was significantly higher in the group with the most favorable outcome.

TABLE 1 Demographic and clinical characteristics of 132 patients with GGE

	JME N = 87	Other GGE N = 45 ^a
Female/Male	50/37 (57%/43%)	26/19 (58%/42%)
Mean age at inclusion (y)	25.7 (SD = 6.7)	23.5 (SD = 6.7)
Mean age at first seizure	14.8 (SD = 2.8) ^b	12.1 (SD = 5.2)
Mean epilepsy duration (y)	10.9 (SD = 6.4)	11.4 (SD = 8.1)
History of absence seizures	28 (32%)	30 (67%)
History of myoclonic jerks	87 (100%)	12 (27%)
History of GTCS	80 (92%)	39 (87%)
Off antiepileptic drugs >1 y	18 (21%)	7 (16%)
Never started antiepileptic drug treatment	4 (5%)	1 (2%)
Monotherapy	56 (64%)	34 (76%)
Polytherapy	13 (15%)	4 (9%)
No GTCS within the last year	69 (79%)	37 (82%)
No seizures within the last year ^c	28 (32%)	26 (58%)

SD, standard deviation.

^a8 childhood absence epilepsy, 22 juvenile absence epilepsy, 15 epilepsy with GTCS only.

^bMean age at first motor seizure. Nine patients (10%) had CAE evolving to JME.

^cIncluding GTCS, absences, and myoclonic jerks.

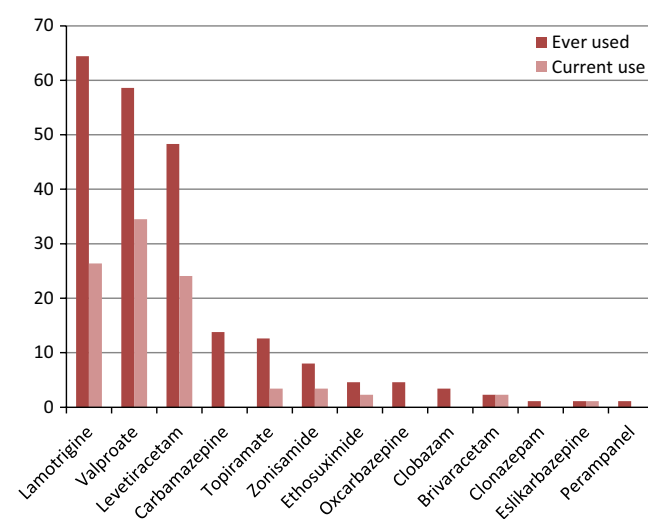


FIGURE 1 History of antiepileptic drug use in 87 patients with JME. Numbers are in percent

4 | DISCUSSION

To the best of our knowledge, we present the largest study assessing AED withdrawal in JME, reflecting the regular clinical setting of

TABLE 2 Clinical and psychosocial factors in 87 patients with JME in relation to self-withdrawal of AED treatment

	Self-withdrawal of AEDs N = 26	No self-withdrawal of AEDs N = 61
Female/male	16/10 (62%/38%)	34/27 (56%/44%)
Age at inclusion (y)	26.9 (SD = 6.5)	25.2 (SD = 6.8)
Epilepsy duration (y)	11.8 (SD = 5.9)	10.6 (SD = 6.6)
Polytherapy	1 (4%)	12 (20%)
Antiepileptic drugs, previous or present		
Lamotrigine	19 (73%)	37 (61%)
Levetiracetam	8 (31%)	32 (53%)
Topiramate	5 (19%)	6 (10%)
Valproate	18 (69%)	34 (56%)
Zonisamide	2 (8%)	5 (8%)
Adverse effects of antiepileptic drugs ^b	11 (42%)	27 (44%)
Identifiable seizure trigger factors	25 (96%)	55 (90%)
Currently employed or studying	21 (81%)	47 (77%)
High school degree	18 (69%)	35 (57%)
Ever in need of psychiatric health care	15 (58%)	32 (53%)
Parent with psychosocial difficulties ^{b,c}	15 (58%)	15 (25%)
Current quality of life ^e	6.9 (SD = 2.6)	7.3 (SD = 7.3)
Total BIS score	66.8 (SD = 9.5)	64.4 (SD = 10.3)
Total HADS score	10.8 (SD = 6.0)	11.9 (SD = 6.9)
Depression subscore	4.0 (SD = 3.5)	3.8 (SD = 3.0)
Anxiety subscore	6.9 (SD = 4.0)	8.1 (SD = 4.5)

Statistically significant findings are marked in bold.

SD, standard deviation.

^bRash on lamotrigine, and/or mood change on levetiracetam, and/or weight gain on valproate.

^cAddiction or violent behavior, as reported by the patient.

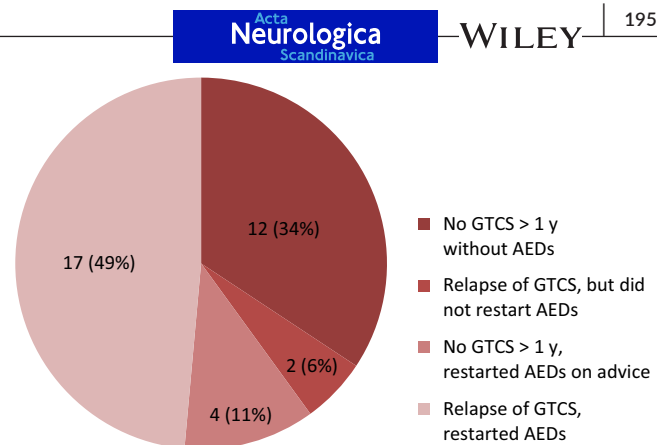
^dStatistically significant ($P \leq 0.05$).

^eMeasured by a visual analogue scale (1-10).

a regional department of neurology. In spite of the general advice against treatment withdrawal in JME, 40% of the included patients had discontinued AED treatment. Self-withdrawal was common, and more so in JME than in other types of GGE. Nevertheless, 1/3 of those who discontinued AED therapy remained free from GTCS and did not restart AEDs.

4.1 | Self-withdrawal of antiepileptic drugs

Considering previous studies strongly discouraging AED withdrawal in JME,⁵⁻⁷ it may not be surprising that most of the JME patients opting for treatment withdrawal did so without consulting a doctor. The treating neurologist may not have provided information about the possibility of AED withdrawal, or even advised against it. Several

**FIGURE 2** Antiepileptic drug withdrawal in 35 patients with JME

authors state that there could be a JME-specific, risk-taking behavioral pattern related to executive dysfunction, as a consequence of deficits in thalamo-frontal-cortical neuronal circuits.²²⁻²⁴ Self-withdrawal of AED may be considered a type of risk-taking behavior, as the chance of GTCS recurrence would be highly present. Thus, our findings are in support of the hypothesis of a JME-specific behavioral pattern. Furthermore, it has been demonstrated that the neuropsychological deficits delineated in JME are present in otherwise healthy siblings as well.³⁰ Bearing in mind that frontal lobe dysfunction and risk-taking behavior may lead to psychosocial difficulties like drug abuse and criminal behavior, it is possible that such problems may be found to a greater extent in families affected by JME.^{28,31} We find it remarkable that the single factor standing out as over-represented in the self-withdrawal JME group was having a parent with psychosocial difficulties like addiction or violent behavior. It is likely that parents with poor psychosocial outcome would have difficulties adhering to medical advice. However, mean age at seizure onset in the present study was 15 years. Consequently, the decision of self-withdrawal was most likely made by the patients themselves. All in all, if it is the case that JME families are prone to issues related to frontal lobe deficit, the follow-up and care for these patients and their families require special attention, particularly when it comes to treatment adherence, as the present study demonstrates.

Adverse effects, on the other hand, did not seem to be more common in the self-withdrawal group, which is surprising. Adverse effects are normally closely linked to non-adherence and sometimes influence quality of life to a greater extent than seizures per se.³²⁻³⁴ However, adverse effects were not systematically assessed by means of standardized tools in the present study. Troublesome weight gain caused by valproate, irritability and aggression caused by levetiracetam, and rash caused by lamotrigine were inquired about specifically. Other common and sometimes disabling adverse effects of AEDs were not included, such as dizziness, headaches, and nausea.

Another interesting aspect was that more than 60% of self-withdrawal happened within 2 years of the last GTCS. This could reflect distrust in the information about a possible need for lifelong treatment, and it could reflect willingness to take a risk. Nevertheless, nearly 60% of those who self-withdrew AEDs had no relapse of

	No AEDs and no GTCS > 1 y N = 12	Recurrence of GTCS and/or restart of AEDs N = 23
Female/male	8/4 (67%/33%)	13/10 (57%/43%)
Age at inclusion (y)	28.1 (SD = 6.8)	24.3 (SD = 6.7)
Epilepsy duration (y)	11.9 (SD = 6.1)	11.1 (SD = 6.5)
Age at first motor seizure (y) ^b	16.2 (SD = 2.6)	13.2 (SD = 2.4)
< 2 y since last GTCS at time of withdrawal	7 (58%)	13 (57%)
2-5 y since last GTCS at time of withdrawal	3 (25%)	8 (35%)
> 5 y since last GTCS at time of withdrawal	2 (17%)	2 (9%)
Photoparoxysmal response in EEG	4 (33%)	15 (65%)
Polyspikes in EEG	8 (67%)	11 (48%)
JME evolved from CAE	2 (17%)	3 (13%)
Three seizure types ^c	2 (17%)	6 (26%)
Self-withdrawal	11 (92%)	15 (65%)
Currently employed or studying	9 (75%)	21 (91%)
High school degree	8 (67%)	16 (70%)
Ever in need of psychiatric health care	8 (67%)	11 (48%)
Total BIS score	66.1 (SD = 11.3)	64.7 (SD = 8.1)
Total HADS score	11.4 (SD = 6.2)	10.2 (SD = 6.5)
Depression subscore	3.2 (SD = 2.5)	4.1 (SD = 3.7)
Anxiety subscore	8.3 (SD = 4.7)	6.1 (SD = 3.6)

Statistically significant findings are marked in bold.

SD, standard deviation.

^bStatistically significant ($P \leq 0.05$)

^cAbsences, GTCS, and myoclonic jerks

TABLE 3 Clinical and psychosocial factors in the 35 JME patients who withdrew AED treatment in relation to outcome

GTCS and did not restart AED therapy. In fact, the rate of GTCS was lower in those who did not use AEDs than in those who were on medication, although not significantly. Myoclonic jerks on the other hand seemed to be more common in those who did not use AEDs. It is possible that patients who discontinued AED therapy had a milder type of JME with rare GTCS and that those who experienced more GTCS or a more active type of epilepsy were more motivated to continue AEDs. It seems that as long as GTCS do not recur, patients accept myoclonic jerks in order to avoid taking AEDs.

4.2 | Outcome of antiepileptic drug withdrawal

More than half of the JME patients withdrawing medication in the present study experienced GTCS relapse. Efforts have been made to identify predictors of treatment resistance in JME, and we found that age at first motor seizure was significantly lower in the group experiencing relapse and/or restarting AED therapy. This is in line with the conclusions of a recent meta-analysis, stating that low age at seizure onset is a predictor of refractory JME. They also found

that absence seizures, three seizure types, JME evolving from CAE, and psychiatric comorbidities were predictors of refractoriness.³⁵ In the present study, neither three seizure types, nor need of psychiatric health care, nor JME evolving from CAE was over-represented among those who experienced relapse of GTCS and/or restarted AEDs. The meta-analysis did not find photoparoxysmal response (PPR) to be a predictor of treatment-resistant JME.³⁵ In the present study, photoparoxysmal response was more common in the group experiencing recurrence of GTCS and/or restarting AED therapy, but not significantly. Interestingly, a German study found a significant association between seizure recurrence and PPR in JME patients withdrawing medication, while there was no association between treatment resistance and PPR. The study was small, however ($N = 31$).¹⁶ Nevertheless, PPR as a possible predictor of seizure recurrence following AED withdrawal in JME may be an interesting and useful lead to follow in future studies. A patient with PPR may attain seizure control easily, but possibly has an increased risk of recurrence if treatment is withdrawn. Consequently, larger studies of the outcomes of AED withdrawal in JME, and not just seizure

outcome in general, are needed in order to help predicting the risks at AED withdrawal in the individual JME patient.

4.3 | Methodological considerations

A strength of the study is the recruitment of patients through a systematic search of medical records containing close to all patients with epilepsy in our region. Furthermore, a considerable participation rate was reached by offering home visits to those who were unable to come to the hospital. Thus, selection bias was reduced. The retrospective review of medical records made us able to identify those who no longer attended regular follow-up at the hospital, an essential point when it comes to exploring treatment adherence. That the interviews were independent of regular clinical follow-up is a strength as well. This probably contributed to more reliable answers when it came to adherence to treatment and medical advice.

A limitation of the present study was that observation time of seizure outcome was short. It is possible that several of the patients who had no GTCS within the last year could have a GTCS relapse at a later stage, especially if frequency of GTCS was low in the first place. However, in the hospital-based design of the study, people older than 40 years were excluded in order to avoid bias toward treatment-resistant epilepsy. Consequently, a different study design and recruitment method would be needed (ie, a follow-up study) in order to provide information about 5- and 10-year seizure outcome.

Another methodological consideration is the restricted sample size, making sophisticated statistical methods like logistic regression less suitable. Regression models would have been valuable in the assessment of potential predictors of self-withdrawal and the outcomes of such withdrawal. Nevertheless, we identified factors of interest, which could be further explored in future studies of larger sample sizes.

The lack of a systematic assessment of adverse effects is also a limitation. We sought to minimize the number of questionnaires in order to increase participation and reliability of results, considering that attention and ability to concentrate may be reduced in JME.²² Thus, only a few adverse effects considered relevant for the drugs most commonly used in the treatment of JME were inquired about. Furthermore, investigating other aspects of non-adherence in JME would be important, as the present study evaluated self-withdrawal only. Executive dysfunction could lead to challenges related to organizing and maintaining a stable intake of AEDs. Consequently, a study of non-intentional non-adherence in JME would be of interest, for instance, by means of therapeutic drug monitoring (TDM).³⁶

5 | CONCLUSION

Withdrawal of AEDs is common in JME, and self-withdrawal in particular. Self-withdrawal may be related to troublesome conditions at home, and special attention must be paid to the follow-up of people with JME and their families, particularly when it comes to

treatment adherence. Some may continue without AEDs and remain free from GTCS, however. A higher age at first motor seizure may indicate a more favorable outcome.

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CONFLICT OF INTEREST

Marte Syvertsen served on the advisory board of Eisai's epilepsy educational program and received speaker honoraria from Eisai. The remaining authors have no conflict of interests.

AUTHORS' CONTRIBUTION

Marte Syvertsen designed the study, had a major role in the acquisition and analyses of the data, and drafted the manuscript. Ida Fløgstad had a major role in the acquisition of data, assisted in analysis of the data, and revised the manuscript. Ulla Enger had a major role in the acquisition of data. Cecilie Johannessen Landmark had a major role in analyses and interpretation of the data and in revising the manuscript. Jeanette Koht designed and supervised the study, and revised the manuscript.

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